

Angus Addlesee

Senior Research Scientist · Research @ PolyAI

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Profile

Senior Research Scientist at PolyAI, training the in-house models behind its voice agents: models built for the hard constraints of live phone calls, where latency, accuracy, and reliability all matter at once. Previously Applied Scientist in Amazon Alexa's Frontier AI Modelling Lab, working on Alexa+ across LLM routing and optimisation, reward modelling for evaluation and training, and high-throughput evaluation infrastructure. PhD in Artificial Intelligence focused on accessible, human-centred dialogue systems for people with dementia. 50+ international talks, EACL Best Demo Award, and work featured in TIME Magazine.

Experience

- 2026–Present **Senior Research Scientist, PolyAI** ■ *Research*, Cambridge, UK
- Training PolyAI's in-house models for voice agents, built for the hard constraints of live phone calls where latency, accuracy, and reliability all matter at once.
 - Work spans the Raven family of voice-native LLMs and audio-native dialogue models.
- 2024–2026 **Applied Scientist, Amazon Alexa+** ■ *Frontier AI Modelling Lab*, Cambridge, UK
- LLM routing and optimisation for Alexa+ (production models triggered for 100% of traffic).
 - Lead the development of reward models for model evaluation and training, moving to run on 100% of production traffic.
 - Built a high-throughput system simulator enabling offline evaluation of counterfactual tool-calling, used for router training and model checkpoint evaluation against traffic signals.
 - On-call rotas: diagnose and resolve Alexa+ production issues under pressure.
 - Active interviewer: conducted 25+ technical interviews for the team.
- 2022–2024 **Research Associate, Heriot-Watt University**, Edinburgh, UK
- EU SPRING H2020 project: conversational AI and robotics for dementia care.
 - Led the conversational AI and dialogue system development for a multi-party social robot deployed in a Parisian memory clinic.
 - EACL 2024 Best Demo Award; project featured in TIME Magazine.
- Mar–Aug 2022 **Applied Scientist Intern, Amazon Alexa**, Cambridge, UK
- Research internship during PhD on voice assistant accessibility and partial-question understanding. Submitted three papers (INTERSPEECH, CUI, AMLC).
- 2019–2022 **Machine Learning Engineer & Researcher, Wallscope**, Edinburgh, UK
- NLP and ML projects for NHS Scotland and Scottish Government (real-time discharge-letter parsing into SNOMED-grounded knowledge graphs feeding a live GP UI; child-protection analytics). Concurrent with PhD.
- 2018–2019 **Machine Learning Engineer, Wallscope**, Edinburgh, UK
- Information extraction from unstructured data into knowledge graphs for healthcare and public sector clients.
- 2017–2018 **Data Scientist, Wallscope**, Edinburgh, UK
- Data-driven NLP solutions for healthcare and government services.

Education

- 2019–2024 **PhD in Artificial Intelligence, Heriot-Watt University**, Edinburgh, UK
- Thesis: *Incremental multi-party conversational AI for people with dementia*. Funded by The Data Lab.
- 2016–2018 **MSc in Artificial Intelligence with Speech and Multimodal Interaction, Heriot-Watt University**, Edinburgh, UK, *Distinction*
- Thesis: *An Exploration into Open-domain Question Answering for Conversational Agents*.

2013–2016 **BSc (Hons) in Mathematics**, *Heriot-Watt University*, Edinburgh, UK
Dissertation: *Markov Chains: Modelling Stochastic Processes and Exploring the Mathematics Behind these Models*.

Awards

- 2024 **EACL Best Demo Award**, European Chapter of the Association for Computational Linguistics. *Multi-party Multimodal Conversations Between Patients, Their Companions, and a Social Robot in a Hospital Memory Clinic*.
- 2022 **Societal Impact Award**, Heriot-Watt University.
- 2019 **Industrial Doctorate Funding Award**, The Data Lab Scotland.

Speaking & Leadership

- 50+ Talks International talks at venues including **Harvard, USC, Carnegie Mellon, KTH, BBC, NatWest**, plus conferences across Europe, North America, and the Middle East (Boulder, Boston, Dublin, Eindhoven, Glasgow, Leipzig, Los Angeles, Ljubljana, Malta, Montreal, Paris, Prague, Riga, Stockholm, Turin, Abu Dhabi, and more).
- Co-founder **European & African Conversational AI Summits** (since 2020). The first European event hosted 1,200 attendees with 46 speakers; the first African event hosted 1,000 attendees with 44 speakers, the first event of its kind in Africa.
- SIGdial Officer Special Interest Group on Discourse and Dialogue.
- Conference Organising SIGdial 2022 (Edinburgh); YRRSDS 2021, 2022; SLiDInG 7 (with the Scottish Government); ICMI 2020 (supported).
- Supervision 60+ MSc students supervised across multiple cohorts on conversational AI, accessibility, and human-computer interaction projects.

Selected Publications

- 2025 *Socially Pertinent Robots in Gerontological Healthcare*. Alameda-Pineda et al. International Journal of Social Robotics.
- 2024 *Multi-party Multimodal Conversations Between Patients, Their Companions, and a Social Robot in a Hospital Memory Clinic*. Addlesee et al. EACL (**Best Demo Award**).
- 2024 *You Have Interrupted Me Again!: Making Voice Assistants More Dementia-Friendly With Incremental Clarification*. Addlesee & Eshghi. Frontiers in Dementia.
- 2024 *Clarifying Completions: Evaluating How LLMs Respond to Incomplete Questions*. Addlesee, Lemon & Eshghi. LREC-COLING.
- 2023 *Multi-party Goal Tracking with LLMs: Comparing Pre-training, Fine-tuning, and Prompt Engineering*. Addlesee et al. SIGdial.
- 2024 **PhD Thesis**: *Incremental Multi-party Conversational AI for People with Dementia*. Heriot-Watt University.
Full list at Google Scholar.

Skills & Interests

- Technical LLM Routing & Optimisation, Large Language Models, Reward Modelling, Conversational AI, Real-Time Systems, Speech & Dialogue, Knowledge Graphs, Production ML.
- Research Human–AI Interaction, Accessibility, Responsible AI, Multimodal Interaction, Healthcare AI, Multi-party Dialogue.
- Languages English (native).